Software Requirements Specification (SRS)

SmartCart AI – AI-Powered Grocery Comparison and Recommendation System

1. Introduction

Project Name: SmartCart AI
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Purpose: To develop a web application that helps customers compare grocery products across nearby shops and make informed purchasing decisions using price, quantity, customer ratings, and distance.

2. Overall Description

SmartCart AI is a web-based grocery comparison and recommendation platform with two primary user roles: customers and shop owners. Customers can compare grocery listings, view recommendations, manage their carts, and review order summaries. Shop owners can manage product listings, prices, quantities, and stock availability. The system uses Supabase for authentication, database management, and data storage, with a recommendation engine that evaluates multiple product and location factors.

3. Functional Requirements

- **FR1 – Authentication:** The system shall support user registration and login with role-based access.
- **FR2 – Product Comparison:** Customers shall be able to compare grocery products across shops based on price, quantity, ratings, and distance.
- **FR3 – Recommendations:** The system shall recommend suitable products by considering value for money, customer ratings, and proximity.
- **FR4 – Cart Management:** Customers shall be able to add, update, and remove products from their cart.
- **FR5 – Shop Management:** Shop owners shall be able to add and update product listings, prices, quantities, and stock.
- **FR6 – Order Summary:** The system shall display selected products and their corresponding prices in a consolidated summary.

4. Non-Functional Requirements

- **Performance:** The application should load pages and retrieve product information within a reasonable time.
- **Security:** User authentication and database access shall be protected using appropriate access controls and Supabase Row Level Security policies.
- **Usability:** The interface shall be simple, intuitive, and responsive across desktop and mobile devices.
- **Reliability:** The system should handle invalid inputs and database errors gracefully.

- **Maintainability:** The application should use modular components and clearly separated responsibilities.

5. Interfaces

- **User Interface:** Responsive web interface for customers and shop owners.
- **Software Interface:** React, TypeScript, Vite, Supabase, and PostgreSQL.
- **Database Interface:** Supabase APIs for authentication, product data, shop information, carts, and related records.
- **Network Interface:** HTTPS communication between the client application and Supabase services.

6. Constraints

- The application requires an internet connection.
- Product availability, prices, and ratings depend on the accuracy of stored data.
- Location-based comparisons depend on valid customer and shop coordinates.
- Recommendation quality depends on the implemented scoring logic and available data.
- Access to customer and shop-owner data must follow configured authorization policies.

7. Appendices

Acronyms: SRS – Software Requirements Specification; AI – Artificial Intelligence; UI – User Interface; DB – Database.

Technology References: React, TypeScript, Vite, Supabase, PostgreSQL, and Vercel.

Future Enhancements: Online payments, delivery tracking, real-time stock updates, and improved personalized recommendations.